$$n = \frac{\sin\left(\frac{A+\delta}{2}\right)}{\sin\left(\frac{A}{2}\right)} \quad (6.90)$$

For small angles of A and δ,

$$\sin\left(\frac{A+\delta}{2}\right) \approx \left(\frac{A+\delta}{2}\right)$$

$$\sin\left(\frac{A}{2}\right) \approx \left(\frac{A}{2}\right)$$

$$\therefore n = \frac{\left(\frac{A+\delta}{2}\right)}{\left(\frac{A}{2}\right)} = \frac{A+\delta}{A} = 1 + \frac{\delta}{A}$$

On further simplifying, $\frac{\delta}{A} = n - 1$

$$\delta = (n-1)A \quad (6.91)$$

When white light enters the prism, the deviation is different for different colours. Thus, the refractive index is also different for different colours.

Let δ_V, δ_R are the angles of minimum deviation for violet and red colour. Let n_V and n_R be the refractive indices for the violet and red colour respectively.

For Violet colour, $\delta_V = (n_V - 1)A$ (6.92)

For Red colour, $\delta_R = (n_R - 1)A$ (6.93)

As, angle of minimum deviation for violet colour δ_V is greater than angle of minimum deviation for red colour δ_R, the refractive index for violet colour n_V is greater than the refractive index for red colour n_R.

Subtracting δ_R from δ_V we get,

$$\delta_V - \delta_R = (n_V - n_R)A \quad (6.94)$$

The angular separation between the two extreme colours (violet and red) in the spectrum ($\delta_V - \delta_R$) is called the *angular dispersion*.

If we take δ as the angle of minimum deviation for any mean colour (green or yellow) and n the corresponding refractive index. Then,

$$\delta = (n - 1)A \quad (6.95)$$

Dispersive power ω is defined as the ratio of the angular dispersion for the extreme colours to the deviation for any middle colour. Dispersive power is the ability of the material of the prism to produce dispersion.

$$\omega = \frac{\text{angular dispersion}}{\text{middle deviation}} = \frac{\delta_V - \delta_R}{\delta} \quad (6.96)$$

Substituting for $(\delta_V - \delta_R)$ and (δ),

$$\omega = \frac{(n_V - n_R)}{(n-1)} \quad (6.97)$$

The dispersive power is a dimensionless and unitless quantity. The dispersive power is always positive. The dispersive power of a prism depends only on the nature of material of the prism and it is independent of the angle of the prism.

EXAMPLE 6.22

Find the dispersive power of a prism if the refractive indices of flint glass for red, green and violet colours are 1.613, 1.620 and 1.632 respectively.

Solution

Given, n_V=1.632; n_R=1.613; n_G=1.620

Equation for dispersive power is,

$$\omega = \frac{(n_V - n_R)}{(n_G - 1)}$$

Substituting the values,

$$\omega = \frac{1.632 - 1.613}{1.620 - 1} = \frac{0.019}{0.620} = 0.0306$$

The dispersive power of the prism is, $\omega = 0.0306$

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